

APIs & Automation Closure Report

- **Project Name:** ParaBank Website

- **Introduction:**

This Test Closure Report summarizes the activities, results, and outcomes of the API Testing and Automation Testing phases performed for the ParaBank Website. The objective was to validate the reliability, functionality, and consistency of the backend services, and to automate high-priority functional scenarios to reduce regression effort.

- **Testing Scope:**

- **API Testing Scope:**

- API testing covered the validation of major service endpoints related to:

- Authentication
 - Accounts Overview
 - Account Details
 - Fund Transfers
 - Bill Payment
 - Loan Request
 - ATM Services (Deposit, Withdraw, Check Balance)

- **Automation Scope:**

- Automation scripts were created for the highest-priority functional cases including:

- Authentication (Login, Registration)
 - Logout
 - Forgot Login Info
 - Funds Transfer
 - Bill Payment
 - Viewing Accounts Overview
 - Opening Account Details
 - Database actions
 - New accounts rules updates

- **Test Activities Performed:**

- **API Testing Activities:**

- Identified and analyzed API endpoints.
 - Designed and executed API test cases using Postman.

- **Automation Testing Activities:**

- Selected automation tools: Selenium WebDriver.
 - Built the automation framework structure.
 - Executed automated test cases on Chrome browser.

- **API Testing Results:**

APIs Tested	Total Test Cases	Passed	Blocked	Failed
18	63	51	1	11

- **Automation Testing Results:**

Automated Test Cases	Passed	Failed
38	37	1

- **Critical Defects Found:**

- Negative amounts accepted.
- Wrong date format accepted.
- Negative transfer amount accepted.

- **Risks & Mitigations**

- Some API endpoints still lack consistent error messages.
- Limited test data availability may affect automated runs.

- **Exit Criteria Status:**

Exit Criteria	Status
All high-priority API test cases executed	Completed
All automated scripts executed successfully	Completed
Test summary reports prepared	Completed

- **Conclusion:**

- All exit criteria have been met.
- API Testing and Automation Testing for the ParaBank application were completed successfully.
- The backend services demonstrated stable functionality with an 80% API pass rate, while the automated suite achieved a 97% pass rate across core business workflows.